

The study of phase relationships plays an important role in the understanding of the properties of materials. Phase diagrams are maps that give the relationships between phases *in equilibrium* in a system as a function of temperature, pressure and composition. The phases as determined from the microstructure of a material may or may not correspond to the equilibrium phases indicated by the phase diagram. In fact, the thermal treatment given to a material often results in phases other than those in the diagram. This fact, however, does not diminish the importance of the study of phase diagrams as a first step towards gaining an insight into the control of microstructure.

Units

Quantity	SI units		Other units
	<i>Unit</i>	<i>Symbol</i>	
Temperature	kelvin	K	°C, °F
Pressure	pascal or newton per square metre	Pa or N m ⁻²	atm, psi, kbar

The compositions given in the text are weight %, unless stated otherwise.

7.1 The Phase Rule

The phase rule enunciated by Gibbs has a simple form:

$$F = C - P + 2 \quad (7.1)$$

where F is the degrees of freedom, C is the number of components and P is the number of phases in equilibrium in a system. Let us examine each of these terms in some detail.

In the classical definition, a *phase* is a physically distinct, chemically homogeneous and mechanically separable region of a system. The various states of aggregation of matter, namely, the gaseous, the liquid and the solid states, form separate phases. The gaseous state is always a single phase, as the atoms (or molecules) in the gas are mixed at the atomic (or molecular) level. A liquid solution is also a single phase. For example, if salt is dissolved in water, the water molecules, the sodium ions and the chlorine ions are mixed at the atomic level in the solution. A liquid mixture (e.g. oil and water), on the other hand, forms two separate phases as there is no mixing at the molecular level. In the solid state, different chemical compositions and different crystal structures are possible so that a solid may consist of several phases. For the same composition, different crystal structures represent different phases. A solid solution has the atoms mixed at the atomic level within the unit cell and is therefore a single phase.

The *components* of a system may be elements, ions or compounds. They refer to the independent chemical species that comprise the system. In the ice–water–steam system, the component is H_2O . In the Cu–Ni system, the elements Cu and Ni are the components, whereas in the Al_2O_3 – Cr_2O_3 system, the two oxides can be taken to be components. In the Fe–C system, iron and graphite can be the components, but it may be often convenient to choose iron and iron carbide (Fe_3C) as the components.

The *variables* of a system include the two external parameters, *temperature and pressure*. Within the system, there are variables that specify the compositions of the phases present. Compositions are expressed as weight (or atom %) so that the number of variables required to specify completely the composition of a phase is $(C - 1)$, where C is the number of components in the system. Knowing the percentage of $(C - 1)$ components automatically fixes the percentage of the last component. If there are P phases in a system, the total number of composition variables is $P(C - 1)$. Including the two external variables (pressure and temperature), the total number of variables is $P(C - 1) + 2$. The number of *independent variables* among these gives the degrees of freedom F . Clearly, this number cannot be more than the total number of variables:

$$F = C - P + 2 \leq P(C - 1) + 2 \quad (7.2)$$

It is easy to see from Eq. (7.2) that, when only one phase is present in a system, the degrees of freedom are equal to the total variables. As the number of phases increases, the degrees of freedom decrease. The degree of freedom cannot be less than zero so that we have an upper limit to the number of phases that can exist in equilibrium in a given system.

Example 7.1 What are the degrees of freedom of a system of two components when the number of phases is one, two, three, and so on?

Solution For $C = 2$, we can list the total variables and the degrees of freedom as follows:

No. of phases	Total variables $P(C - 1) + 2$	Degrees of freedom $C - P + 2$
1	3	3
2	4	2
3	5	1
4	6	0

The system cannot have more than four phases in equilibrium.

Phase diagrams are classified on the basis of the number of components in the system. Single-component systems have unary diagrams, two-component systems have binary diagrams, three-component systems give rise to ternary diagrams, and so on.

7.2 Single-component Systems

In single-component systems, there is no composition variable and the only other variables are temperature and pressure. The phase diagram for iron is shown in Fig. 7.1. Pressure is plotted on the x-axis and temperature on the y-axis. The pressure axis starts from 0.1 MPa but can be extended to lower pressures. In phase diagrams, solid phases are denoted by Greek alphabets α , β , γ , etc.

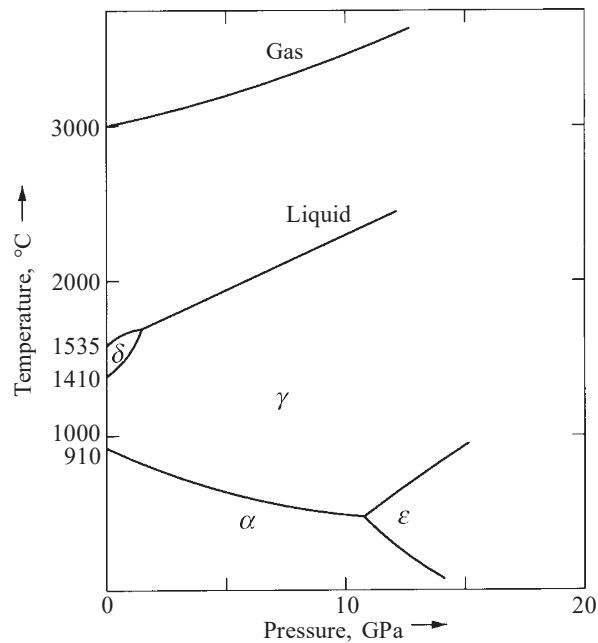


Fig. 7.1 Pressure-temperature diagram for the one-component system of iron.

Consider first the single-phase regions on the diagram such as gas, liquid and several crystal forms of iron. From the phase rule [Eq. (7.1)] for $C = 1$ and $P = 1$, we have $F = 2$. Therefore, in a single-phase region, both temperature and pressure can be varied independently within the limits prescribed by the boundaries of the region.

When two phases are in equilibrium, $F = 1$, that is, either temperature or pressure can be varied independently, *but not both*. Two-phase equilibrium exists *along the phase boundaries* of Fig. 7.1. When we are at a point on the phase boundary if we change the temperature by an arbitrary amount and thereby exercise a degree of freedom in order to preserve the two-phase equilibrium, we need to change the pressure by such an amount that we return to a point on the phase boundary.

Three-phase equilibrium exists at points on the phase diagram, where three phase boundaries meet. Two such points are shown in Fig. 7.1. Such points are called *triple points*. Here, $F = 0$; neither pressure nor temperature can be varied arbitrarily. Three phases will co-exist at only one particular combination of pressure and temperature. If we alter pressure or temperature from the fixed triple-point value, one or two of the phases will disappear.

The equilibrium crystal form of iron at ambient temperature and pressure is BCC (α). On heating to 910°C, α changes over to the FCC (γ) form*. On heating to 1410°C, the γ iron changes over to the BCC (δ) form. As the pressure is increased, the $\alpha \rightarrow \gamma$ transition temperature is lowered, whereas the $\gamma \rightarrow \delta$ transition temperature is increased. Recent high pressure experiments have shown that, when a pressure of about 15 GPa (~150 000 atm) is applied at room temperature, the BCC iron transforms to the HCP (ϵ) phase, as illustrated in Fig. 7.1.

7.3 Binary Phase Diagrams

Two-component systems have binary phase diagrams. Apart from temperature and pressure, we have one composition variable for each of the phases in equilibrium. We then need a three-dimensional diagram to plot the variations in pressure, temperature and composition. In order to simplify the representation of the phase relationships on paper, binary phase diagrams are usually drawn at atmospheric pressure, showing variations in temperature and composition only. Pressure changes often produce no significant effect on the equilibrium and, therefore, it is customary to ignore the pressure variable and the vapour phase. In such cases, one of the variables has been arbitrarily omitted and the phase rule for the condensed phases (solid and liquid phases only) is written in a modified form as

$$F = C - P + 1 \quad (7.3)$$

*Originally, it was believed that a β form of iron also existed, which was later disproved. To avoid confusion, the original naming sequence of the solid phases has been retained with the β phase deleted.

Example 7.2 At atmospheric pressure (pressure arbitrarily chosen), a material of unknown composition shows four phases in equilibrium at 987 K. What is the minimum number of components in the system?

Solution As pressure is arbitrarily chosen, we can use the modified form of the phase rule as given in Eq. (7.3). The minimum number of components corresponds to the minimum degree of freedom, which is zero. Taking $F = 0$, we get

$$0 = C - 4 + 1, \quad C = 3.$$

The simplest binary phase diagram is obtained for a system exhibiting complete liquid solubility as well as solid solubility. The two components dissolve in each other in all proportions both in the liquid and the solid states. Clearly, the two components must have the same crystal structure besides satisfying the other Hume Rothery's conditions for extensive solid solubility, see Sec. 5.3. Cu–Ni, Ag–Au, Ge–Si and Al_2O_3 – Cr_2O_3 are examples of such systems.

Figure 7.2 shows the phase diagram of Al_2O_3 – Cr_2O_3 . Pure Al_2O_3 and pure Cr_2O_3 form the left and the right end of the composition axis. They are arranged in alphabetical order from left to right. The composition is read as per cent of Cr_2O_3 starting from 0% at left and going to 100% at the right end. Alternatively, the composition can be read as per cent of Al_2O_3 from right to left. Temperature is shown along the y-axis.

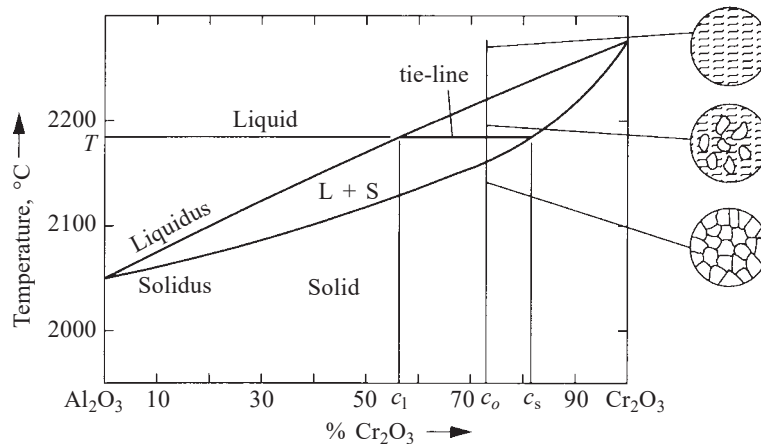


Fig. 7.2 The Al_2O_3 – Cr_2O_3 phase diagram. Microstructural changes in an overall composition c_0 are sketched on the right.

There are only two phases on the phase diagram, the liquid and the solid phases. The single-phase regions are separated by a two-phase region ($L + S$), where both liquid and solid co-exist. In all binary phase diagrams, a two-phase region separates single-phase regions, as given by the “1-2-1” rule. As we move from a single-phase region (1), we cross into a two-phase region (2), and then

again into a single-phase region (1)^{**}. The phase boundary between the liquid and the two-phase region is called the *liquidus*. The boundary between the solid and the two-phase region is called the *solidus*.

When only one phase is present, the composition axis gives the composition of that phase directly. When two phases are present, the compositions of the phases are not the same. They should be read according to the following convention: At the temperature of interest T , a horizontal line called the *tie-line* is drawn as shown in Fig. 7.2. *The points of intersection of the tie-line with the liquidus and the solidus give, respectively, the liquid and the solid compositions, c_l and c_s , which are in equilibrium with each other.* Thus, in Fig. 7.2, at 2180°C, for

an overall composition $c_o = 73\% \text{ Cr}_2\text{O}_3$ (27% Al_2O_3),

we have

The liquid composition $c_l = 57\% \text{ Cr}_2\text{O}_3$ (43% Al_2O_3), and

The solid composition $c_s = 82\% \text{ Cr}_2\text{O}_3$ (18% Al_2O_3).

The phase rule can be applied to this phase diagram, using the modified form given in Eq. (7.3). For the single-phase region (liquid or solid), from Eq. (7.3), $F = 2 - 1 + 1 = 2$. So, both temperature and the composition of the phase can be independently varied (within limits). In the two-phase region, $F = 2 - 2 + 1 = 1$.

Here, we have three variables:

- (i) Temperature
- (ii) Composition of the liquid phase
- (iii) Composition of the solid phase

As $F = 1$, only one of these three is independent. If we arbitrarily choose the temperature, the compositions of the two phases are automatically fixed and are given by the ends of the tie-line drawn at that temperature. If we specify the composition of one of the phases arbitrarily, the temperature and the composition of the other phase are automatically fixed. There is no three-phase equilibrium in systems exhibiting complete solid solubility.

Many pairs of elements and compounds are unlikely to satisfy the conditions for complete solid solubility. For instance, the size difference between two atoms or ions can be appreciably more than 15%, as the table of atomic and ionic radii indicates (see back inside cover of the book). Similarly, the other conditions for extensive solubility may not be satisfied. The solid solubility is therefore limited in a number of binary systems. But it is never zero. However unfavourable the conditions for solid solubility are, a very small quantity of any component will always dissolve in another component as this increases the configurational entropy and lowers the free energy of the crystal, recall Problem 2.10. The solubility may be so small that, for all practical purposes, only the pure component may be shown on the phase diagram.

When solid solubility is limited and the melting points of the components are not vastly different, a *eutectic phase diagram* usually results. As an example,

^{**}This is true, except when the phase boundary is a horizontal line, corresponding to an invariant temperature.

the Pb–Sn phase diagram is shown in Fig. 7.3. As there is complete liquid solubility, the liquid phase extends over all compositions above the melting temperatures of the components. The solid phase at the left end is the lead-rich

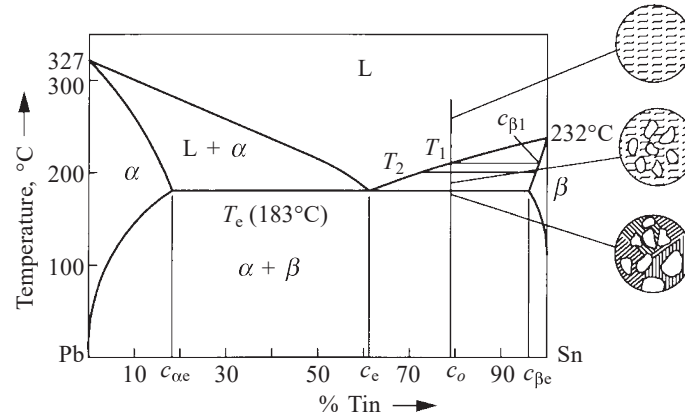


Fig. 7.3 The lead–tin phase diagram. Microstructural changes for composition c_o are sketched.

α which dissolves only a limited amount of tin. This solubility decreases with decreasing temperature. This limit of the solid solubility is indicated by the phase boundary between α and $\alpha + \beta$, called *solvus*. The solid solution phase at the right end is the tin-rich β , with only a very small quantity of lead dissolved in it. The phase boundaries on this diagram are as follows:

- Liquidus I: boundary between L and $L + \alpha$ regions
- Liquidus II: boundary between L and $L + \beta$ regions
- Solidus I: boundary between α and $L + \alpha$ regions
- Solidus II: boundary between β and $L + \beta$ regions
- Solvus I: boundary between α and $\alpha + \beta$ regions
- Solvus II: boundary between β and $\alpha + \beta$ regions

The three two-phase regions are separated by a horizontal line corresponding to the temperature T_e called the *eutectic temperature*. Below the eutectic temperature, the material is fully solid for all compositions. The composition which remains fully liquid up to the eutectic temperature during cooling is called the *eutectic composition* c_e . At the eutectic temperature, the following *eutectic reaction* takes place:



Here, cooling refers to heat being extracted from the system at the eutectic temperature, and heating refers to heat being added to the system. As there is some heat evolution or absorption during the reaction, it is possible to add or subtract heat at constant temperature. The eutectic horizontal is used as a tie-line, the ends of which give the compositions of the α and β phases at this temperature. Summarizing the eutectic characteristics of the Pb–Sn system, we have

eutectic temperature $T_e = 183^\circ\text{C}$,
 composition of liquid $c_e = 62\% \text{ Sn (38\% Pb)}$,
 composition of α , $c_{\alpha e} = 18\% \text{ Sn (82\% Pb)}$, and
 composition of β , $c_{\beta e} = 97\% \text{ Sn (3\% Pb)}$.

The phase rule is readily applied in the single-phase and the two-phase regions of the diagram, on the same lines as discussed for the $\text{Al}_2\text{O}_3\text{--Cr}_2\text{O}_3$ system. At the eutectic temperature T_e , three phases are in equilibrium. From Eq. (7.3), $F = 2 - 3 + 1 = 0$. The eutectic temperature T_e and the compositions of the three phases, c_e , $c_{\alpha e}$ and $c_{\beta e}$ are all fixed and none of them can be varied arbitrarily. On slightly increasing the temperature above T_e , either one or both of α and β phases would disappear. On slight decrease of temperature below T_e , the liquid phase would transform as per reaction (7.4) to a mixture of α and β . To denote the zero degree of freedom, the eutectic reaction is called an *invariant reaction*. The eutectic temperature is known as an *invariant temperature*.

A similar invariant reaction occurring entirely in the solid state, where the liquid phase is replaced by a third solid phase γ , is called a *eutectoid reaction*:



The corresponding invariant temperature is called the *eutectoid temperature*.

When the melting points of the components are vastly different from each other, a *peritectic phase diagram* may be formed. As an example, the Ag–Pt phase diagram is shown in Fig. 7.4. The melting points of the components differ

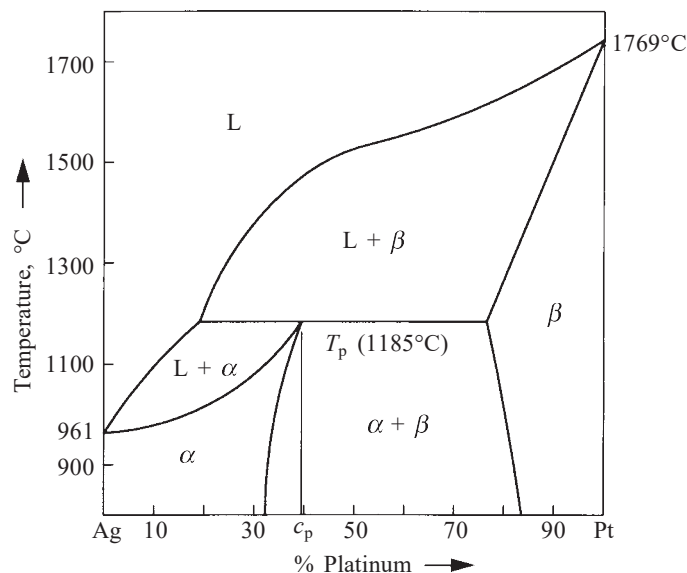


Fig. 7.4 The silver–platinum phase diagram.

by more than 800°C. At the peritectic temperature T_p , the following invariant reaction occurs:



In contrast to the eutectic reaction, the peritectic reaction is between a liquid and a solid phase to yield on cooling a single solid phase. The peritectic composition c_p is the composition of the product solid phase. The peritectic horizontal is the tie-line which defines the composition of the liquid phase and the β phase at the peritectic temperature.

Analogous to the eutectoid reaction, we have a peritectoid reaction, where two solids react on cooling to produce a third solid phase:



The invariant reactions that we have considered up till now are summarized in Table 7.1.

TABLE 7.1
Invariant Reactions

Name of the reaction	Details of the reaction	Phase boundaries at the invariant line
Eutectic	cooling $\rightarrow$ $L \leftrightarrow \alpha + \beta$ $\leftarrow$ heating	
Eutectoid	cooling $\rightarrow$ $\gamma \leftrightarrow \alpha + \beta$ $\leftarrow$ heating	
Peritectic	cooling $\rightarrow$ $L + \beta \leftrightarrow \alpha$ $\leftarrow$ heating	
Peritectoid	cooling $\rightarrow$ $\gamma + \beta \leftrightarrow \alpha$ $\leftarrow$ heating	

Binary diagrams have other invariant reactions, such as monotectic and syntectic reactions. In addition, many binary systems have compounds at certain ratios of the components, e.g., Mg_2Pb in the Mg–Pb system. Some of these compounds exist over a range of composition rather than at a fixed value. Further details about binary phase diagrams and systems with more than two components can be found in the references listed at the end of this chapter.

7.3 Microstructural Changes during Cooling

The microstructural changes that occur on cooling an overall composition c_o in the $\text{Al}_2\text{O}_3\text{--Cr}_2\text{O}_3$ diagram are illustrated in Fig. 7.2. Above the liquidus, the material is a single phase (liquid). On cooling, we cross the liquidus and enter the two-phase region. At temperature T , solid crystals of composition c_s co-exist with liquid of composition c_l . On further cooling, we cross the solidus and the material is now fully crystallized into a polycrystalline solid of composition c_o .

In the eutectic system of Fig. 7.3, let us consider the cooling of an overall composition c_o . At temperature T_1 , solidification starts with a very small quantity of β crystals of composition $c_{\beta 1}$ coming out of the liquid. At a lower temperature T_2 , the amount of β phase has increased. Its composition is now different and is given by the tie-line drawn at this temperature. Just above the eutectic temperature, the quantity of β crystals has further increased, their composition now being $c_{\beta e}$. The microstructure of the alloy at this stage would be composed of β crystals floating in the liquid of eutectic composition, as shown in Fig. 7.3. On cooling just below the eutectic temperature, the liquid transforms according to the eutectic reaction (7.4) to yield a fine mixture of thin plate-like crystals of α and β , see Fig. 7.3. The microstructure below the eutectic temperature is re-sketched in Fig. 7.5. The large equiaxed crystals of *proeutectic* β , that is, β

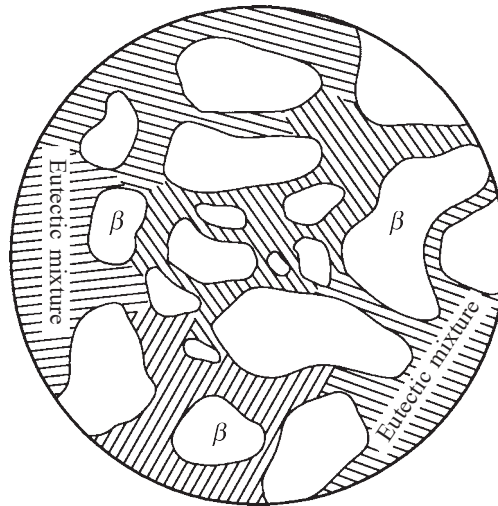


Fig. 7.5 Microstructure of a hypereutectic Pb–Sn alloy, showing the proeutectic β crystals in a matrix of the eutectic mixture (schematic).

formed before the eutectic reaction took place, are clearly seen. The matrix in which they are embedded is the eutectic mixture consisting of thin, parallel plates of alternate α and β crystals. Since the eutectic region is clearly distinguishable as a separate entity under the microscope, it is called a *micro-constituent*. It should not, however, be called the eutectic phase as it is not a single phase.

If the overall composition of the alloy is c_e , the entire solidification takes place at the eutectic temperature and the microstructure in this case has only the eutectic mixture, as illustrated in Fig. 7.6. If the overall composition is to the left

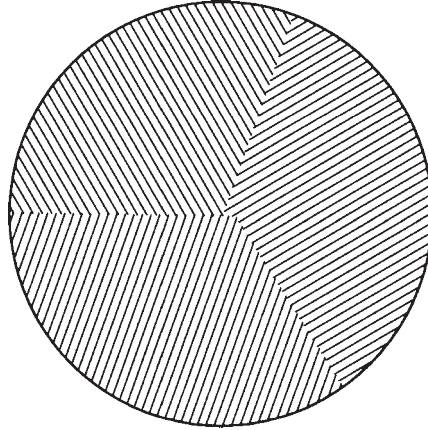


Fig. 7.6 Microstructure of a Pb-Sn alloy of eutectic composition (schematic).

of the eutectic composition, then the microstructure consists of large equiaxed crystals of proeutectic α that form prior to the eutectic reaction. These α crystals get embedded in the eutectic matrix that forms at the eutectic temperature. Alloy compositions to the left of the eutectic mixture are called *hypoeutectic alloys*, and those to the right are called *hypereutectic alloys*.

What we have described above corresponds to an extremely slow cooling so that the concerned phases have the equilibrium composition as given by the tie-line at every temperature. This means that the composition of the equiaxed β crystals of the hypereutectic alloy of Fig. 7.3 should change continuously from $c_{\beta 1}$ to $c_{\beta e}$, as we cool from T_1 to T_e . Atoms must move across the liquid-crystal boundary (from or to the liquid) to bring about these compositional adjustments. Atom movements in the solid are so slow that these adjustments are rarely complete. At ordinary cooling rates, the crystals show a gradation of composition from $c_{\beta 1}$ at the centre of the crystals, where solidification first started, to $c_{\beta e}$ at the peripheral rims of the crystals which solidified just above the eutectic temperature. This compositional inhomogeneity is called *coring*. Coring can be identified by the contrast a cored crystal shows on chemical etching.

Sometimes crystals grow *preferentially* along certain crystallographic directions. Frequently, in cubic crystals, the three mutually perpendicular $\langle 100 \rangle$ directions are the preferred directions of growth. During the early stages of solidification, the shape of the crystal is like a fir tree, with arms protruding out in three directions, as seen in Fig. 7.7. This structure is called the *dendritic structure*. The liquid between the dendritic arms undergoes solidification subsequently. The compositional differences caused by the coring effect enable the identification of the dendrites even after complete solidification.

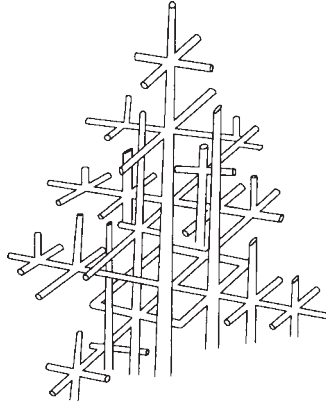


Fig. 7.7 The dendritic structure.

7.5 The Lever Rule

As already described, the compositions of the two co-existing phases of a binary system are given by the tie-line rule. For all overall compositions that lie on the tie-line, the compositions of the two phases remain the same, c_s and c_l in Fig. 7.2. A little reflection will show that this will be possible only if the relative amounts of the co-existing phases change, as the overall composition is varied along the tie-line. *Lever rule* derived from mass balance gives the relative amounts of the co-existing phases. It is applied as follows: *The tie-line is treated as a lever arm, with the fulcrum at the overall composition.* For the arm to be horizontal, the weight to be hung at each end must be proportional to the arm length on the other side of the fulcrum. The “weight” at each end corresponds to the amount of the phase at that end. At temperature T and overall composition c_o in Fig. 7.2, the relative amounts of the liquid and the solid phases are determined as follows. Figure 7.8 illustrates the procedure. Expressing the

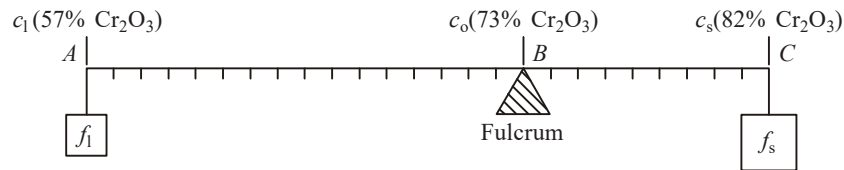


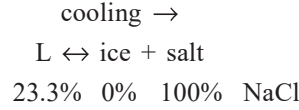
Fig. 7.8 Illustration of the lever rule.

weight fractions of liquid and solid as f_l and f_s , we obtain

$$f_l = \frac{CB}{CA} = \frac{c_s - c_o}{c_s - c_l} = \frac{82 - 73}{82 - 57} = 0.36 \quad (7.8)$$

$$f_s = \frac{BA}{CA} = \frac{c_o - c_l}{c_s - c_l} = \frac{73 - 57}{82 - 57} = 0.64 \quad (7.9)$$

Example 7.3 The H_2O – NaCl system has the following eutectic reaction occurring at -21°C :



How much pure water can be extracted from sea water (3.5% NaCl) by cooling to -20.9°C ?

Solution -20.9°C is just above the eutectic temperature of -21°C . Here, liquid of eutectic composition (23.3% NaCl) is in equilibrium with pure ice (0% NaCl). The ice crystals can be separated from the liquid and melted into water. With the fulcrum at the overall composition of sea water (3.5% NaCl) and the lever arm extending to 23.3% NaCl on one side to 0% on the other side, we obtain

$$f_{\text{ice}} = (23.3 - 3.5)/(23.3 - 0) = 19.8/23.3 = 0.85$$

In the above examples, only two phases are present. The lever rule cannot be applied at the eutectic or the peritectic temperature, where there are three phases in equilibrium and an isothermal reaction, changing the relative amounts of the phases, can occur. It can be applied just above or just below the invariant temperature. By using the lever rule, it is possible to estimate

- (i) the fraction of a proeutectic phase,
- (ii) the fraction of the eutectic mixture, and
- (iii) the fraction of a phase that forms part of the eutectic mixture.

Referring to Fig. 7.3, just below the eutectic temperature, for the overall composition c_o , the fraction of proeutectic β , $f_{\text{pro } \beta}$, is determined as follows: The fulcrum of the lever is positioned at the overall composition. One end of the lever arm ends at the phase boundary corresponding to β , that is, at the composition $c_{\beta e}$. The other end of the lever arm extends up to the average composition of the eutectic mixture c_e . We can then write

$$f_{\text{pro } \beta} = \frac{c_o - c_e}{c_{\beta e} - c_e} \quad (7.10)$$

The fraction of the eutectic mixture is given by

$$f_{\text{eut}} = \frac{c_{\beta e} - c_o}{c_{\beta e} - c_e} \quad (7.11)$$

In order to determine the total β , which is the sum of the proeutectic β and the β in the eutectic mixture, the fulcrum as before remains at the overall composition. The ends of the lever arm, however, extend up to $c_{\beta e}$ on the β side and up to $c_{\alpha e}$ on the α side.

$$f_{\text{total } \beta} = \frac{c_o - c_{\alpha e}}{c_{\beta e} - c_{\alpha e}} \quad (7.12)$$

Example 7.4 What is the fraction of β that forms part of the eutectic mixture in the alloy of overall composition of c_o in Fig. 7.3?

Solution The fraction of the β phase that forms part of the eutectic mixture is equal to the total β in the material minus the proeutectic β . From Eqs. (7.10) and (7.12),

$$\begin{aligned} f_{\text{eut } \beta} &= f_{\text{total } \beta} - f_{\text{pro } \beta} \\ &= \frac{c_o - c_{\alpha e}}{c_{\beta e} - c_{\alpha e}} - \frac{c_o - c_e}{c_{\beta e} - c_e} \end{aligned}$$

This fraction can also be expressed as the product of two fractions as follows:

$$\begin{aligned} f_{\text{eut } \beta} &= f_{\text{eut in this alloy}} \times f_{\beta \text{ in eut alloy}} \\ &= \frac{c_{\beta e} - c_o}{c_{\beta e} - c_e} \times \frac{c_e - c_{\alpha e}}{c_{\beta e} - c_{\alpha e}} \end{aligned}$$

It is left as an exercise for the student to show that both these expressions for fraction of eutectic β are equal.

7.6 Summary of Phase Diagram Rules

As a review, the rules pertaining to phase diagrams are summarized as follows:

7.6.1 The Phase Rule

The Gibbs phase rule states that

$$F = C - P + 2$$

where F is the degrees of freedom, P the number of phases, and C the number of components. A phase is a region of a system with the same structure and uniform composition and differs from other regions of the system either in structure and/or composition. The component refers to the individual chemical species. The degrees of freedom are the number of independent variables associated with the equilibrium. It is equal to the total variables, when there is only one phase in the system. It decreases as the number of phases increases. The maximum number of phases in equilibrium in a system corresponds to $F = 0$.

7.6.2 The Tie-Line Rule

The tie-line rule is applied to determine the compositions of two co-existing phases in a binary phase diagram. The tie-line is a horizontal line drawn at the temperature of interest within the two-phase region. If, for example, a liquid phase and a solid phase co-exist at a temperature T , the intersection of the

tie-line drawn at that temperature with the liquidus gives the composition of the liquid, and the intersection with the solidus gives the composition of the solid. The tie-line rule is not concerned with the fraction of the co-existing phases. It can be applied only in the two-phase region. Note that, for all overall compositions that lie on the tie-line, the compositions of the two co-existing phases remain the same. There is change only in their relative amounts, which is determined by the lever rule.

7.6.3 The Lever Rule

The lever rule gives the fractions of two co-existing phases. The tie-line at the temperature of interest is treated as a lever arm, with the fulcrum at the overall composition. For the lever arm to be horizontal, the weight hung at each end must be proportional to the arm length on the other side of the fulcrum. The “weight” at each end corresponds to the amount of the phase at that end.

It should be noted that the tie-line rule gives the composition of the co-existing phases, whereas the lever rule gives the fractions (or amounts) of the phases. In a binary system of A and B, if a liquid of 35% A (65% B) is co-existing with a solid of 75% A (25% B), for an overall composition of 40% A, the fraction of the liquid is given by

$$f_l = \frac{75 - 40}{75 - 35} = 0.875$$

The fraction of the solid $f_s = 1 - f_l = 1 - 0.875 = 0.125$.

Stating the tie-line rule and the lever rule together, for the above example, we get

$$\begin{aligned} \text{fraction of liquid with 35\% A} &= 0.875 \\ \text{fraction of solid with 75\% A} &= 0.125 \\ 0.875 \times 35 + 0.125 \times 75 &= 40\% \text{ A} \\ &= \text{overall composition of the alloy} \end{aligned}$$

The lever rule cannot be applied at an invariant temperature, where three phases are in equilibrium. It can, however, be applied just above or below the invariant line. The lever rule can be used to calculate the fraction of a proeutectic phase, the fraction of a eutectic mixture or the fraction of a phase that forms part of the eutectic mixture. In such calculations, the fulcrum is always at the overall composition. If the fraction of a eutectic mixture is to be calculated, one end of the lever arm must end at the average composition of the eutectic mixture, that is, at the eutectic composition.

7.6.4 The 1-2-1 Rule

The 1-2-1 rule is an aid to label the phase fields in a binary phase diagram. As we move from a single-phase region (1), we cross into a two-phase region (2), and then again into a single-phase region (1). The 1-2-1 rule is not applicable across an invariant horizontal line.

Note that the boundaries of a two-phase region always meet at a point on reaching a pure component. This is consistent with the fact that a pure component undergoes a phase change such as crystallization at a constant temperature.

7.7 Some Typical Phase Diagrams

Phase diagrams may look somewhat complicated if there is more than one invariant reaction in the system. As examples, we consider a few more typical phase diagrams.

7.7.1 Magnesia–Alumina System

The $\text{MgO}-\text{Al}_2\text{O}_3$ phase diagram is shown in Fig. 7.9. It has two eutectic reactions, and an intermediate phase, $\text{MgO} \cdot \text{Al}_2\text{O}_3$. There is very little terminal

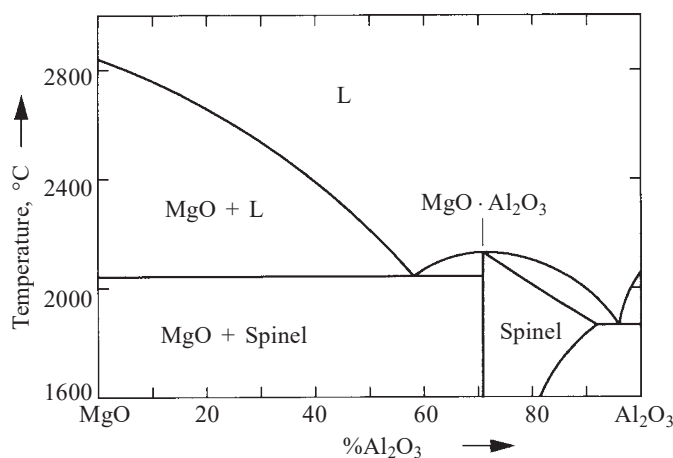


Fig. 7.9 The $\text{MgO}-\text{Al}_2\text{O}_3$ phase diagram.

solubilities, that is, Al_2O_3 dissolves only a negligible amount of MgO and vice versa. So, for all practical purposes, the pure end phases are the same as components. This is in contrast to the $\text{Pb}-\text{Sn}$ system, where we have Pb and Sn as the two components and α and β as the two phases near the pure ends, see Fig. 7.3. The $\text{MgO}-\text{Al}_2\text{O}_3$ system can also be considered as two separate phase diagrams, the components for one being MgO and $\text{MgO} \cdot \text{Al}_2\text{O}_3$, and for the other being $\text{MgO} \cdot \text{Al}_2\text{O}_3$ and Al_2O_3 .

Note that a small quantity of MgO in Al_2O_3 lowers its melting point appreciably, even though MgO has a very high melting point. Such a phenomenon can cause problems when oxides are used as high temperature refractories. High quality pure refractories with a high softening temperature can be ruined due to the presence of impurities that lower their melting point.

7.7.2 Copper–Zinc System

Commercially important Cu–Zn alloys, *brasses*, have a composition up to about 40% Zn. The Cu–Zn phase diagram from 0 to 60% Zn is shown in Fig. 7.10a. Zinc melts at 419°C and copper at 1083°C. This large difference in the melting points results in a number of peritectic reactions in the system, two of which fall in the range shown in Fig. 7.10a. The phases present in brass at room temperature as a function of zinc content are listed as follows:

α	$\alpha + \beta$	β
0–35% Zn	35–46.6% Zn	46.6–50.6% Zn

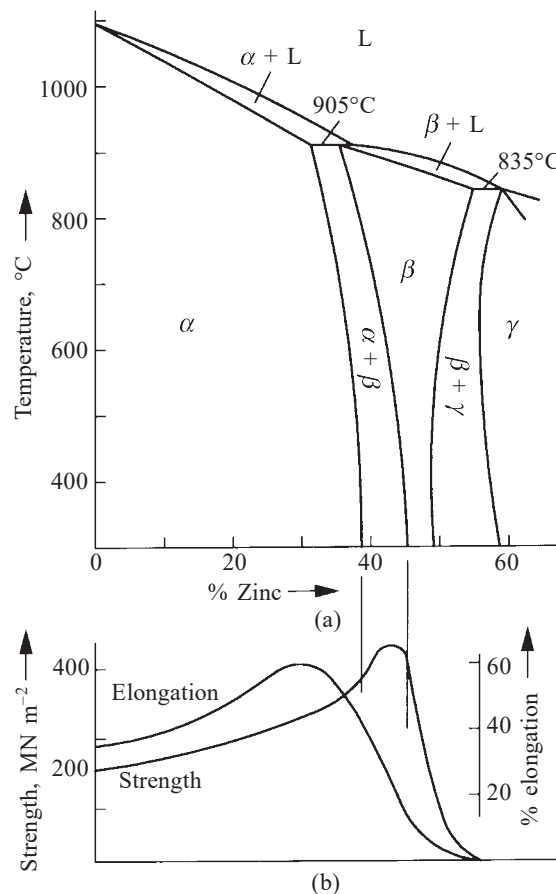


Fig. 7.10 (a) The copper–zinc phase diagram up to 60% zinc; (b) the variation in strength and % elongation with zinc content.

Zinc is cheaper than copper and, therefore, brasses are more economical to use than pure copper. In addition, the mechanical properties of copper such as tensile strength and % elongation improve with the addition of zinc, see

Fig. 7.10b. Brasses are widely used in a variety of applications. Where electrical conductivity is an important consideration, pure copper is used.

7.7.3 Iron–Iron-carbide System

Fe and Fe_3C (iron carbide) are the components in the Fe– Fe_3C phase diagram. It is also possible to have a phase diagram with Fe and C (graphite) as the components. Graphite is more stable than Fe_3C . So, the Fe– Fe_3C diagram can be considered to be a metastable phase diagram. Most steels contain only iron carbide and not graphite.

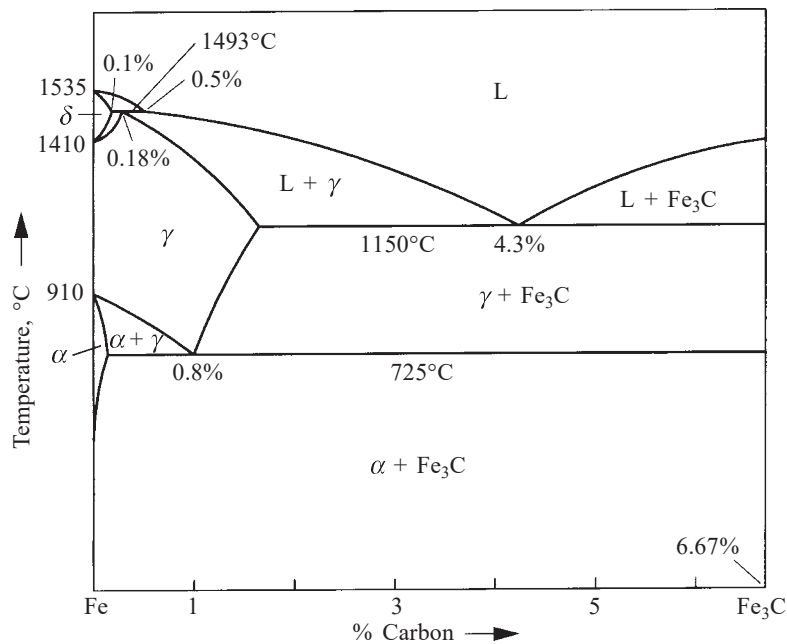


Fig. 7.11 The Iron–Iron-Carbide phase diagram.

In Fig. 7.11, the composition is plotted on the x -axis as weight per cent of carbon. Pure iron melts at 1535°C . A peritectic invariant reaction occurs at 1493°C with the peritectic composition at 0.18% carbon:



The product phase γ is called *austenite*. A eutectic reaction occurs at 1150°C with the eutectic composition at 4.3% carbon:



A eutectoid reaction occurs at 725°C with the eutectoid composition at 0.8% carbon:



Here, austenite (γ) decomposes into two phases—ferrite (α) and cementite (Fe_3C).

Fe–C alloys containing 0–1.4% carbon are called *steels*. The approximate range of carbon content of mild steels is 0–0.3%, for medium carbon steels it is 0.3–0.6%, and for high carbon steels, it is 0.8–1.4%. Here, we will consider the microstructures of slowly cooled steels of different carbon content. When the carbon content is negligible, the structure of steel is essentially polycrystalline α (ferrite), which is the interstitial solid solution of carbon in BCC iron.

When the steel has the eutectoid composition of 0.8% carbon, the microstructure consists of alternate layers of thin, parallel plates of α (ferrite) and Fe_3C (cementite). This eutectoidal mixture is called *pearlite*. Pearlite is a microconstituent, and not a single phase. It is so named as it has an iridescent appearance under the microscope, resembling the mother of pearl, see Fig. 7.12.

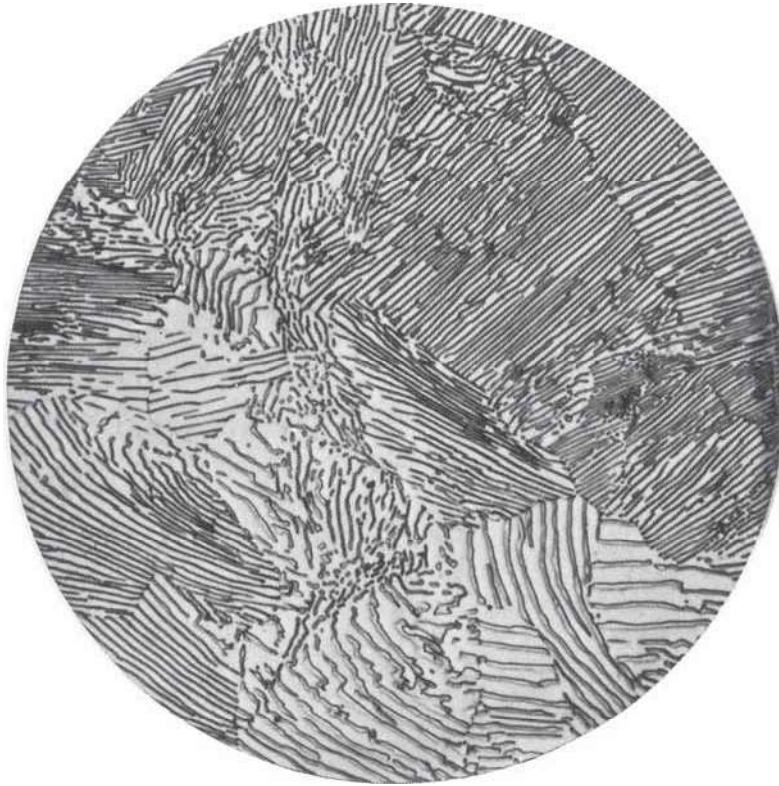


Fig. 7.12 Microstructure of eutectoid steel showing pearlite, magnified 1000 times. (E.C. Bain and H.W. Paxton, *Alloying Elements in Steel*, with permission from ASM International, Materials Park, Ohio.)

Applying the lever rule, with fulcrum at 0.8% carbon and the lever arm extending up to ferrite ($\sim 0.0\%$ carbon) at one end and up to cementite (6.67% carbon) at the other end, we have, in pearlite, the relations

$$f_{\alpha} = (6.67 - 0.8)/(6.67 - 0.0) = 0.88$$

$$f_{\text{Fe}_3\text{C}} = (0.8 - 0.0)/(6.67 - 0.0) = 0.12$$

The approximately-parallel lines in the microstructure indicate the layer arrangement of ferrite and cementite. As the fraction of cementite is less than that of ferrite, the cementite plates are thinner than ferrite in Fig. 7.12. Both the boundaries of a cementite plate are clearly visible in some regions of the microstructure, while in other parts, the cementite plates appear to be just a line, as the two boundaries of a plate are too close to be resolved as separate lines. At lower magnifications or when the pearlite mixture is very fine, even ferrite and cementite may not be resolved as separate platelets and pearlite may appear as just dark regions under the microscope.

Mild steel with 0.2% carbon consists of about 75% of proeutectoid ferrite that forms above the eutectoid temperature and about 25% of pearlite. When the carbon content in the steel is increased, the amount of pearlite increases, until we get the fully pearlitic structure at 0.8% carbon. Beyond 0.8%, high carbon steels contain proeutectoid cementite in addition to pearlite.

Example 7.5 How much proeutectoid ferrite is there in a slowly cooled 0.6% steel? How much eutectoid ferrite is there in the same steel?

Solution Applying lever rule between 0.0 and 0.8% carbon with fulcrum at 0.6% carbon,

$$f_{\text{pro } \alpha} = (0.8 - 0.6)/(0.8 - 0.0) = 0.25$$

$$f_{\text{pearlite}} = 1.0 - 0.25 = 0.75$$

The fraction of ferrite in a eutectoid steel = 0.88

$$f_{\text{eut } \alpha \text{ in this steel}} = 0.75 \times 0.88 = 0.66.$$

In slowly cooled carbon steels, the overall hardness and ductility of the steel are determined by the relative proportions of the soft, ductile ferrite and the hard, brittle cementite. The cementite content increases with increasing carbon content, resulting in an increase of hardness and a decrease of ductility, as we go from low carbon to high carbon steels. Table 7.2 lists typical applications of steels according to carbon content. A rivet (at the top of the list) should have good deformability, and hence has a low carbon content. In contrast, a file (at the bottom of the list) should have high hardness and wear resistance and hence has a high carbon content. Even though we may increase the hardness by increasing the carbon content further, the alloy becomes too brittle to be useful above 1.4% carbon. A rail (in the middle of the table) has 0.6% carbon. It combines some toughness with some hardness and wear resistance.

TABLE 7.2
Some Applications of Plain Carbon Steels

Type of steels	Carbon percentage	Uses
Mild steels	0.0–0.1	Chain links, nails, rivets
	0.1–0.2	Ship hulls, car bodies
	0.2–0.3	Bridges
Medium carbon steels	0.4–0.5	Train axles and wheels
	0.5–0.6	Rails
	0.6–0.7	Chisels
	0.8–0.9	Saw cutters, hammers
High carbon steels*	1.0–1.1	Axes
	1.1–1.2	Razor blades
	1.3–1.4	Scissors, knives, files

*Many of these are used after an appropriate thermal treatment.

Fe–C alloys with more than 2% carbon are called *cast irons*. Consider, for example, the cooling of a cast iron with 3% carbon, see Fig. 7.11. On crossing the liquidus into the (L + γ) region, proeutectic γ (austenite) crystallizes first. On passing through the eutectic temperature, liquid of eutectic composition decomposes to a mixture of austenite and cementite. On further cooling through the eutectoid temperature, the austenite decomposes to pearlite, yielding *white cast iron*. The microstructure of white cast iron consists of cementite and pearlite, see Fig. 7.13. Due to the presence of the large fraction of cementite, white cast iron is very hard and brittle.

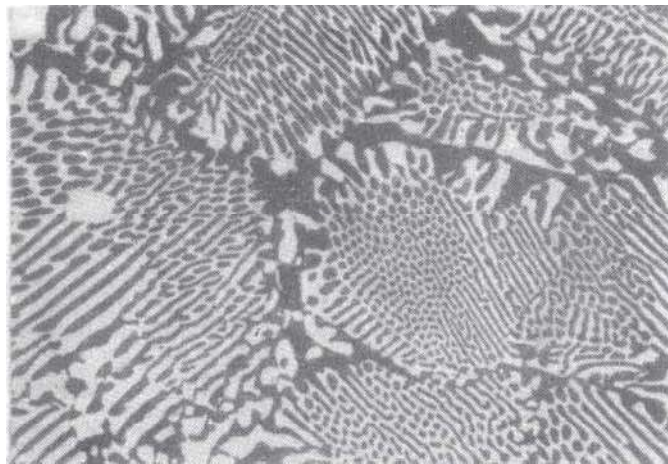


Fig. 7.13 Microstructure of white cast iron, magnified 650 times. White regions are cementite and dark regions are unresolved pearlite. (Y. Lakhtin, *Engineering Physical Metallurgy*, with permission from Mir Publishers, Moscow.)

Depending on the cooling rate and the other alloying elements present in cast iron, the carbon may be present as graphite or cementite. *Gray cast iron* contains graphite in the form of flakes. Slow cooling rates and the presence of silicon promote the formation of graphite. The microstructure of gray cast iron is shown in Fig. 7.14 and consists of graphite flakes in a matrix of ferrite. The



Fig. 7.14 Microstructure of gray cast iron, magnified 140 times. Graphite flakes are embedded in a matrix of ferrite. (A.L. Ruoff, *Introduction to Materials Science*, with permission from Prentice Hall, Inc., Englewood Cliffs, New Jersey.)

graphite flakes are sharp at their tips and act as stress raisers. Due to this, gray cast iron is brittle under tensile loads, in spite of the softness of graphite as compared to the very hard cementite present in white cast iron. The brittleness can be avoided by producing the graphite in the form of spherical nodules (which do not have stress-raising sharp ends), as is done in *malleable cast iron* and SG (*spheroidal graphite*) iron.

Malleable cast iron is produced by heat treating white cast iron for prolonged periods at about 900°C and then cooling it very slowly. The silicon content in the alloy must be 1% or less to ensure that cementite and not flaky graphite forms during solidification. The cementite decomposes to the more stable graphite during the subsequent heat treatment. Graphite appears in the heat-treated microstructure as approximately spherical particles of *temper carbon*. SG iron (also known as *nodular iron*) is produced by making certain alloy additions such as Mg or Ce to molten iron. Here, the silicon content must be about 2.5% to promote graphitization. The alloy additions have the effect of modifying the growth rate of graphite from the melt to be more or less equal in all directions, so that nodules (and not flakes) of graphite are produced. No subsequent heat treatment is required for SG iron.

7.7 Other Applications of Phase Diagrams

One of the important uses of a phase diagram is in the *zone refining* of materials. Consider the schematic phase diagram shown in Fig. 7.15. We need to purify (or refine) A, which has B as the impurity component. Let the starting

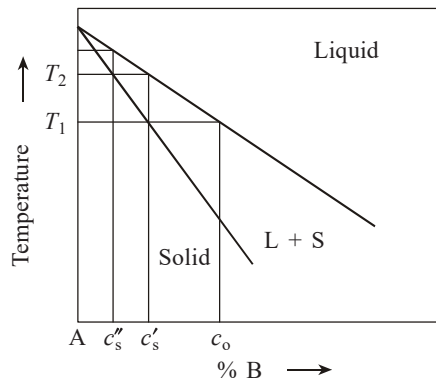


Fig. 7.15 The zone refining process is based on the fact that the solid to crystallize first from a melt is usually purer in A than the liquid.

composition be c_o . This composition starts to solidify at temperature T_1 . Just below this temperature, a small quantity of solid of composition c'_s , purer in A than c_o , separates from the liquid. Imagine that we stop cooling further, throw away the liquid part and remelt the left-over solid. This small quantity of liquid is now purer, as its composition is c'_s . This overall composition will now start to solidify at a higher temperature T_2 . Again, the first solid to separate will have a composition c''_s purer in A than c'_s . By repeating this sequence of operations a few times, we can get very pure A, even though the quantity of the purified material will be extremely small compared to the starting material.

In zone refining, this principle of phase separation is used. The material to be purified is in the form of a long rod. At any time, only a small length of this rod is melted with the aid of an induction coil or an electron beam. The coil or beam is moved slowly from one end to the other end of the rod, continuously solidifying the molten zone and remelting fresh material ahead. Surface tension forces are usually strong enough to hold the molten zone in place without the need for a container, which may contaminate the melt. If the zone is passed across many times, each time in the same direction, the material at the starting end becomes much purer than the rest of the rod. In a typical case, ten passes of the molten zone can reduce the impurity level to as low as 10^{-6} times the initial value. This technique to produce ultra high purity materials has enabled us to achieve the sophistication required in the control of impurity concentration in semiconductor crystals.

The depression of the melting point of a pure solid, when another component goes into solution, has its advantages and disadvantages. The problem of softening and melting of refractories due to impurities was referred

to earlier. Some of the advantages are described below. The eutectic reaction in the iron-carbon alloys facilitates the melting of cast irons at a lower temperature than is required for steels. So, the making of cast iron is easier and more economical. In cold countries, the depression of the freezing point of water by the addition of salt is used to melt the ice on roads. Pb–Sn eutectic alloys are useful as soldering material. Low melting eutectic alloys are used as safety devices in fire fighting equipment and petroleum storage tanks. Gold–silicon eutectic finds uses in the manufacture of semiconductor devices. Closing this section of applications in a lighter vein, a teaspoon made out of a four-component eutectic of Pb, Sn, Bi and Cd melts at 70°C, inside a hot cup of tea!

SUMMARY

1. The phase rule gives the degrees of freedom that are permissible for a system of a given number of components and phases in equilibrium.
2. The pressure-temperature diagrams of one component systems have phase boundaries corresponding to two-phase equilibrium and triple points corresponding to three phase equilibrium.
3. Binary phase diagrams are temperature-composition plots, where the pressure variable is omitted.
4. Eutectic and peritectic systems have a fixed (invariant) temperature for the existence of three-phase equilibrium. Here, the compositions of all the three phases are fixed.
5. The solid state analog of the eutectic and the peritectic reactions are the eutectoid and the peritectoid reactions.
6. Lever rule is a simple rule of mass balance, giving the relative amounts of two phases (or microconstituents) for a given overall composition.
7. Brasses are more economical and have better mechanical properties than copper.
8. The microstructures of slowly cooled steels and cast irons correspond to the phases shown in the phase diagram.
9. The zone refining process is based on the fact that the first solid to crystallize in a two-component system is generally purer than the liquid.

PROBLEMS

- 7.1 Water vapour, ice and water are in equilibrium at 0.01°C and at a pressure of 613 Pa. Which of these phases would disappear, if
- (i) the temperature is decreased, and
 - (ii) the pressure is increased?

- 7.2 In a binary phase diagram (pressure omitted), what is the maximum number of phases that can co-exist for at least one degree of freedom?
- 7.3 Give the number of variables and the degrees of freedom at the peritectic temperature of a binary phase diagram.
- 7.4 Trace the microstructural changes on slow cooling from the liquid state of an alloy of any overall composition that undergoes the peritectic reaction in the Ag–Pt system (Fig. 7.4).
- 7.5 Dendritic growth cannot be identified by an etching technique in pure materials. Why?
- 7.6 Show that, for correct mass balance, the relative amounts of two co-existing phases or microconstituents must be as given by the lever rule.
- 7.7 The following data apply to the binary system of A and B:

Melting point of pure A = 1050°C

Melting point of pure B = 1900°C

At 1250°C, the α solid solution (containing 50% of B), the β solid solution (containing 80% of B) and liquid (containing 30% of B) are in three-phase equilibrium. At room temperature, the maximum solubility of B in the α phase is 30% and the maximum solubility of A in the β phase is 10%.

- (i) Sketch the phase diagram of A and B, showing the phases present in each area of the diagram.
- (ii) What is the reaction that takes place at 1250°C?
- (iii) What are the weight fractions of the phases present in a material of overall composition 75% B, at 1251°C, 1249°C and at room temperature (RT)?

Answer: (iii) At 1251°C, $f_l = 0.1$, $f_\beta = 0.9$;
 at 1249°C, $f_\alpha = 0.167$, $f_\beta = 0.833$; and
 at RT, $f_\alpha = 0.25$, $f_\beta = 0.75$.

- 7.8 In the Pb–Sn system, determine the fraction of β phase in an alloy of 80% Sn at 184°C and 182°C.

Answer: 0.51 and 0.78.

- 7.9 In the microstructure of a Pb–Sn solder alloy, it was found that 88% of the area is occupied by the eutectic constituent and the remaining area by the proeutectic β phase. Assuming that the area fractions are equal to the volume fractions, determine the approximate composition of the alloy. The densities of α and β phases are 10 300 and 7300 kg m⁻³.

Answer: 65.7%.

- 7.10 The phase diagram of a binary system of A and B has a three-phase equilibrium at 250°C, with the compositions of α , liquid and β phases equal to 10%, 55% and 95% B. Just below 250°C, find the compositions at which the proeutectic phase is 1½ times the eutectic mixture.

Answer: 28%, 79%.

- 7.11 Draw the steel region of the Fe–Fe₃C phase diagram and make neat sketches of microstructures expected for four compositions between 0.0 and 1.2% C.

- 7.12 From the Fe–Fe₃C phase diagram, for a 0.2% C steel, name the phases and their fractions at equilibrium at the following temperatures:

- (i) just above 1493°C,
- (ii) just below 1493°C,
- (iii) just above 725°C, and
- (iv) just below 725°C.

Answer: (i) $f_{\delta} = 0.75$, $f_l = 0.25$ (ii) $f_{\gamma} = 0.94$, $f_l = 0.06$

(iii) $f_{\alpha} = 0.77$, $f_{\gamma} = 0.23$ (iv) $f_{\alpha} = 0.97$, $f_{\text{Fe}_3\text{C}} = 0.03$.

- 7.13 What is the fraction of proeutectoid cementite in (i) 1.4% C, (ii) 1.0% C, and (iii) 0.7% C steels?

Answer: (i) 0.10, (ii) 0.034, and (iii) 0.

- 7.14 In cooling from the γ range, both proeutectoid ferrite and proeutectoid cementite form at the austenite grain boundaries. If a structure at room temperature contains 5% of grain boundary phase and 95% of pearlite, what are the possible values for the carbon content? Suggest a method to determine the right composition.

Answer: 1.09% and 0.76%.

- 7.15 In the binary phase diagram (Fig. 7.16), mark the various phase fields. Give all the invariant reactions that occur, stating their names.

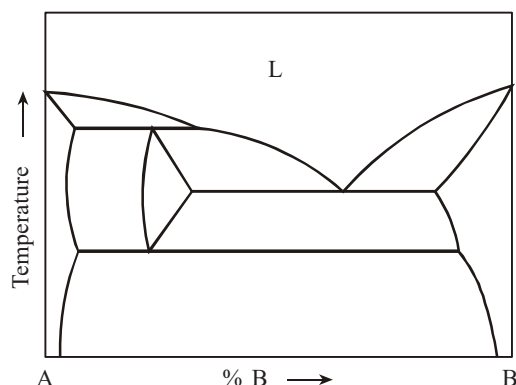


Fig. 7.16 Binary phase diagram of A and B.

7.16 If an impurity raises the melting point of a pure component, how will the zone refining process be affected?

7.17 Derive the degrees of freedom for a system, which has equal number of components and phases.

Answer: 2.

7.18 One solid phase on heating through an invariant temperature becomes two solid phases. Name the invariant reaction. Sketch the phase boundaries near the invariant line.

7.19 In the Pb–Sn system, calculate the alloy composition at which the fraction of total α is $2\frac{1}{2}$ times the fraction of the β phase at 182°C .

Answer: 40.6% Sn.

7.20 For soldering with Pb–Sn alloys, at least 85% of the eutectic mixture is preferred in the microstructure. Determine the composition limits of tin that will satisfy this condition.

Answer: 55.4–67.25%.

7.21 Calculate the fraction of proeutectoid ferrite, eutectoid ferrite and total ferrite in a 0.2% C steel.

Answer: 0.75, 0.22 and 0.97.

7.22 The potassium–sodium binary phase diagram has the following invariant reactions:

	cooling $\rightarrow$
at 6.9°C ,	$\beta + \text{L} \leftrightarrow \text{Na}_2\text{K}$
wt.% Na	99 47 54

	cooling $\rightarrow$
at -12.6°C ,	$\text{L} \leftrightarrow \alpha + \text{Na}_2\text{K}$
wt.% Na	23 3 54

(i) Give the name of each of the above reactions.

(ii) Make an approximate sketch of the phase diagram.

(iii) Find the fraction of Na_2K in a 33 wt.% Na alloy at -12.5°C .

Answer: (iii) 0.32.

7.23 In the Ge–P system, a compound GeP exists. At 725°C , Ge, GeP and liquid of 62 at.% P are in three-phase equilibrium. At 577°C , GeP, P and liquid of 94.5 at.% P are in equilibrium. The melting points of Ge and P are 937 and 593°C respectively. Draw an approximate phase diagram of this system. Write down the invariant reactions, giving their names.

7.24 In the Ti– Ti_2Co system, liquid of 27 wt.% Co, β of 17 wt.% Co and Ti_2Co of 38 wt.% Co are in equilibrium at 1020°C . At 685°C , β of 8 wt.% Co, α of 1 wt.% Co and Ti_2Co are in equilibrium. Pure Ti melts at 1670°C and also undergoes a crystal structure change from β (BCC) to α (HCP) on cooling through 882°C . Ti_2Co is stable up to 1058°C .

- (i) Make an approximate sketch of the phase diagram showing temperatures, compositions and phases.
- (ii) Write down the invariant reactions, giving their names.
- (iii) How much α and Ti_2Co are present in a 6 wt.% Co alloy at room temperature?

Answer: (iii) 0.86, 0.14.

7.25 The data on the gold–lead phase diagram are given below. Melting point of Au = 1063°C, Melting point of lead = 327°C. At 434°C, Au, liquid of 43 at.% Pb and Au_2Pb are in equilibrium. At 253°C, Au_2Pb , liquid of 74 at.% Pb and AuPb_2 are in equilibrium. At 222°C, AuPb_2 , AuPb_3 and liquid of 82 at.% Pb are in equilibrium. At 212°C, AuPb_3 , Pb and liquid of 84 at.% Pb are in equilibrium. Draw the phase diagram on a graph paper using a suitable scale. Write down the invariant reactions, giving the name of each.

7.26 In the Fe–Fe₂Nb system, at 1373°C, δ of 3% Nb, liquid of 12% Nb and ϵ of 27% Nb are in equilibrium. At 1210°C, γ of 1% Nb, δ of 1.5% Nb and ϵ of 27% Nb are in equilibrium. At 961°C, γ of 0.4% Nb, α of 0.7% Nb and ϵ of 27% Nb are in equilibrium. All compositions are in atomic%. The melting point of Fe is 1535°C and of Fe₂Nb (33.3 at.% Nb) is 1627°C. The ϵ phase is stable over a composition range of 27 to 33.3 at.% Nb,

- (i) Draw an approximate phase diagram for this system.
- (ii) Write down the invariant reactions in the system, giving their names.
- (iii) What is the fraction of α in an alloy with 10% Nb at 960°C?

Answer: (iii) 0.65.

7.27 The Al–Si phase diagram is of the simple eutectic type. At 577°C, solid Al with 1.5% Si dissolved in it, solid Si and liquid of 12.5% Si are in equilibrium. An aluminium wire is welded to a silicon substrate. Just below the weld inside the substrate, the microstructure shows 2% of a eutectic-like mixture. What is the composition at this location?

Answer: 98.28% Si.

MULTIPLE CHOICE QUESTIONS

1. The maximum number of co-existing in a C -component system is
 A. $C - P + 2$ B. $P(C - 1)$ C. $F - C + 2$ D. $C + 2$
2. In a single-component system, the maximum number of phases that can coexist in equilibrium is
 A. 2 B. 3 C. 4 D. 5

176 *Multiple Choice Questions*

3. The degree of freedom when ice, water and water vapour co-exist in equilibrium is
A. 1 B. triple pt C. 0 D. -1
4. The degrees of freedom, when FCC iron and BCC iron co-exist in equilibrium, are
A. 2 B. 1 C. 0 D. -1
5. The phase boundary between alpha and (alpha + beta) regions is called
A. liquidus B. solidus C. solvus D. none of these
6. The reaction that yields two solid phases on cooling a single solid phase is called
A. eutectoid B. peritectoid C. eutectic D. congruent
7. If one solid phase splits into two solid phases on heating, the reaction is
A. eutectic B. peritectic C. eutectoid D. peritectoid
8. The reaction that, on heating one solid phase, yields another solid phase plus one liquid phase is called
A. eutectic B. eutectoid C. peritectic D. peritectoid
9. If alpha of 82% B and liquid of 57% B are in equilibrium in an alloy of 73% B, the fraction of liquid is
A. 0.36 B. 0.64 C. 36% B D. 0
10. If the fraction of liquid with 57% B, which is in equilibrium with solid of 82% B, is 0.7, the overall composition is
A. 0.3 B. 74.5% B C. 64.5% B D. 25% B
11. In the eutectic phase diagram of Ag-Cu system, the solubility limit at 500°C of copper is 3% in the Ag-rich phase and of Ag is 2% in the Cu-rich phase. In sterling silver (92.5% Ag -7.5% Cu), the per cent of copper in the Ag-rich phase at 500°C is
A. 95.26 B. 4.74 C. 3 D. 98
12. The eutectic mixture in a Pb-Sn solder alloy should be 90%. At the eutectic temperature, alpha of 19% Sn, liquid of 62% Sn and beta of 97% Sn are in equilibrium. The possible compositions of the solder alloy are
A. 57.7% Sn B. 61% Sn C. 65.5% Sn D. 66.3% Sn
13. The fraction of pearlite in a 0.55% C steel is
A. 0.55 B. 0.31 C. 0.69 D. 0
14. At 30°C, hot chocolate (liquid) with 35% chocolate and 65% vanilla transforms to chocolate ripple (eutectic mixture of vanilla containing 10% chocolate and chocolate containing 5% vanilla). Just below 30°C, the fraction of chocolate ripple in a composition with 45% chocolate is
A. 0.17 B. 0.83 C. 0.41 D. 0.59

15. Zone refining will be more efficient if the ratio of impurity in the solid to that in the liquid is
- A. 0.01 B. 0.1 C. 0.4 D. ~1.0

Answers

- | | | | | |
|-------|----------|-------|-------|-------|
| 1. D | 2. B | 3. C | 4. B | 5. C |
| 6. A | 7. D | 8. C | 9. A | 10. C |
| 11. C | 12. A, C | 13. C | 14. B | 15. A |

Sources for Experimental Data

American Ceramic Society, *Phase Equilibria Diagrams (Phase Diagrams for Ceramists)*, Columbus, Ohio, Vol. I (1964) to Vol. XI (1993).

T.B. Massalski (Ed.), *Binary Alloy Phase Diagrams*, Vols. 1–3, ASM International, Materials Park, Ohio (1990).

Suggestions for Further Reading

W.D. Kingery, H.K. Bowen and D.R. Uhlmann, *Introduction to Ceramics*, Wiley, New York (1976), Chap. 7.

A. Prince, *Alloy Phase Equilibria*, Elsevier, Amsterdam (1966).

F.N. Rhines, *Phase Diagrams in Metallurgy*, McGraw-Hill, New York (1956).